UNIVERSITY OF GRAZ
Department of Mathematics
and Scientific Computing

MAT.B0204UB MATHEMATICAL MODELLING
SUMMER SEMESTER 2025/26

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EXERCISE SHEET 3

Due: 29 April 2026, 10:00

EXERCISE 3.1 (From ex. sheet 2)
Consider the following normal form for $\mu \in \mathbb{R}$ and $\varepsilon \in \mathbb{R}$, $|\varepsilon| \ll 1$,

$$\dot{x} = \mu x - x^3 + \varepsilon. \tag{1}$$

1. Perform the stability and bifurcation analysis of (1).
2. Interpret the results. Namely, interpret the persistence of the bifurcation phenomenon for $\varepsilon \neq 0$, if it occurs. Then, compare with the case $\varepsilon = 0$.

EXERCISE 3.2 *(Discrete Population Model: Cobwebs and Periodic Solutions)*
In this exercise, we study the behavior of the stationary solutions of the scaled discrete logistic growth as we vary the value of the system parameter r using a cobweb diagram. Thus, consider the discrete recursion given by

$$x_{n+1} = r x_n (1 - x_n), \qquad x(0) = x_0 \in (0, 1),$$

where $n \in \mathbb{N}$ and $r > 0$ is a proportionality parameter.

1. Using the following values of x_0 and r, compute (e.g. by hand, MS Excel, Matlab, GeoGebra, Mathematica) the corresponding values of the sequence $\{x_n\}_{n\in\mathbb{N}}$ you obtain from the cobweb plot until a pattern is visible. Plot the points (n, x_n), $n = 0, 1, 2, \ldots$ on the Cartesian plane.
   (a) $x_0 = 0.5$, $r = 0.5$
   (b) $x_0 = 0.3$, $r = 1.5$
   (c) $x_0 = 0.5$, $r = 3.2$

   What is the behaviour of x_n as n increases?
2. Choose any starting point $x_0 \in (0, 1)$ and graph (e.g. Matlab, GeoGebra, Mathematica) a cobweb diagram in the unit square $[0, 1]^2$ for the following values of the parameter r. Plot the points (n, x_n), $n = 0, 1, 2, \ldots$ and discuss the behaviour of x_n as n increases.
   (a) $0 < r < 1$
   (b) $1 < r < 3$
   (c) $3 < r < 1 + \sqrt{6} \approx 3.4494$. Try for $r = 3.1$ and $r = 3.4$ and you should be able to see periodic solutions (period = 2).
   (d) $r > 1 + \sqrt{6} \approx 3.4494$. In this case, check whether the solution of the model has the following properties:
      i. a cycle of period 4 for $r = 3.5$

ii. a cycle of period 8 for $r = 3.55$
iii. a cycle of period 16 for $r = 3.568$
iv. a cycle of period 3 for $r = 3.84$.

Explain what happens when $r = 3.6$ and when $r = 4$.

EXERCISE 3.3
We consider the following system, depending on the parameter $\mu \in \mathbb{R}$,

$$\begin{cases} \dot{x} = -y + \mu x(x^2 + y^2), \\ \dot{y} = x + \mu y(x^2 + y^2). \end{cases} \tag{2}$$

1. Perform the linear stability analysis around the steady state $(0, 0)$. What can you conclude for the associated linear system? What can you conclude for the nonlinear system (2)?

2. Rewrite the system (2) in polar coordinates (r, θ) with $r > 0$ and $\theta \in (-\pi, \pi]$ such that

$$\begin{cases} x = r\cos\theta, \\ y = r\sin\theta, \end{cases} \quad \text{and} \quad \begin{cases} r^2 = x^2 + y^2, \\ x\sin\theta - y\cos\theta = 0. \end{cases}$$

   Hint: use the identities

$$\dot{r} = \dot{x}\cos\theta + \dot{y}\sin\theta \qquad \text{and} \qquad \dot{x}\sin\theta - \dot{y}\cos\theta + \dot{\theta}r = 0.$$

3. Provide a physical interpretation of the dynamical system in polar coordinates (r, θ). What do the trajectories look like for $\mu \in \mathbb{R}$?

4. Compare the linear and the nonlinear stability analyses.